

Course: B. Tech.

Branch : CSE

Semester :XI

Subject Code & Name: Artificial Intelligence-BTCOE603(B)

Max Marks: 60

Date:

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

- | | | |
|---|------------|---|
| A) What is AI? Explain four aspects of AI. | Understand | 6 |
| B) Explain model based agent with example | Understand | 6 |
| C) What is PEAS in AI? Interpret in your own words PEAS description for a Vacuum cleaner? | Understand | 6 |

Q.2 Solve Any Two of the following.

- | | | |
|--|------------|---|
| A) Explain the elements of an agent and list down the characteristics of an intelligent agent. | Remember | 6 |
| B) Explain Depth limited search in detail with suitable example. | Understand | 6 |
| C) Explain A* search. in detail with suitable example. | Understand | 6 |

Q. 3 Solve Any Two of the following.

- | | | |
|---|------------|---|
| A) What is constraint satisfaction problem? How to represent any CSP problem? | Understand | 6 |
| B) What is constraint propagation & local consistency in CSP? | Understand | 6 |
| C) How backtracking is used in CSP? Explain with example | Understand | 6 |

Q.4 Solve Any Two of the following.

- | | | |
|---|------------|---|
| A) What is adversarial search? How to formulate a game as a search problem? | Remember | 6 |
| B) Explain working of minimax algorithm with example. | Understand | 6 |
| C) Explain PEAS description of Wumpus World Problem | Apply | 6 |

Q. 5 Solve Any Two of the following.

- | | | |
|--|-------|---|
| A) Convert following statements using First order logic. | Apply | 6 |
| 1. every man is mortal | | |
| 2. x is greater than y" | | |
| 3. john loves everyone | | |
| 4. Every father loves his child | | |

B)	What is conditional probability and Unconditional Probability, joint probability with example.	Remember	6
C)	Describe conditional Independence in Probability	Understand	6

***** End *****